



Assignment #3_A

SECTION A (2*5=10)

1. What are the functions performed in synthesis phase.
2. What are the goals of Semantic Analysis?
3. What is syntax-directed definition?
4. What is syntax-directed translation?
5. Differentiate between S-Attributed Definition and L-Attributed Definition.

SECTION B (7*3=21 Mark)

1. Differentiate between synthesized translation and inherited translation.
2. What is syntax directed translation (SDD)?
3. Draw the syntax tree and DAG for the following expression: $(a*b)+(c-d)*(a*b)+b$

SECTION C (11*3=33 Mark)

1. Write short note on:
 - a. Synthesized attributes
 - b. Inherited attributes
 - c. Dependency graph
 - d. Evaluation order
 - e. Directed Acyclic Graph (DAG)
2. Construct SDT for given grammar:
 $E \rightarrow E + T, E \rightarrow T, T \rightarrow T * F, T \rightarrow F, F \rightarrow \text{digit}$
Construct Annotated Parse Tree for $5 * 4 + 3$.
3. Construct a DAG for the basic block whose code is given below.
 1. $D := B * C$
 2. $E := A + B$
 3. $B := B * C$
 4. $A := E - D$